Name the force and its source

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “weight” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

weight: the gravitational force on an object

normal force: a support force perpendicular to a contact surface

01 Warm-up: bring back an earlier idea

Velocity changes from -4 to -24 m/s in 4 s, then stays at -24 m/s for 8 s.
Find acceleration and describe the speed trend in each interval.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$.
Use g = 9.8 N/kg unless stated. Normal force is perpendicular to contact; tension follows the string. Name the source.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: Draw the book's force diagram.
Name the force on Earth paired with the book's weight.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add and label force arrows.

Book resting on a level table; draw forces on the book only.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Inventory the interactions

A rope pulls a 5 kg crate to the right across a rough level floor. The crate slides right; the rope is horizontal.
Draw every external force on the crate and name its source. Which magnitudes can you determine using g = 9.8 N/kg?

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

The dot represents the object. Chosen object: crate

Marzano check - circle: ① ② ③

04 Mass and weight in two places

A 3.0 kg instrument is moved from Earth to the Moon.
Use g_Earth = 9.8 N/kg and g_Moon = 1.6 N/kg. Complete the comparison table and explain why mass and weight behave differently.

Sentence starter: Both … . They differ in … because … .

Location	Mass (kg)	Weight (N)
Earth	3	____
Moon	3	____

Marzano check - circle: ① ② ③

05 Normal is not always weight

A rope helps support a 5 kg box on a level floor. Its upward force component is 12 N, and the box has no vertical acceleration.
Use g = 9.8 N/kg. Complete the vertical force diagram and find the normal force.
Explain why it is less than mg.

Sentence starter: Both … . They differ in … because … .



Add the floor's normal force separately.

Marzano check - circle: ① ② ③

Name the force and its source

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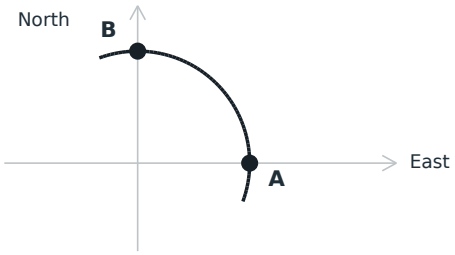
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A puck follows a circle at a steady 5 m/s. A student calls it equilibrium.
Draw velocity at two points and correct the claim.
Use the curve for the turning case; show how velocity changes.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

07 Exit check

A 2.0 kg object hangs at rest from a vertical string.
Name the two forces, their sources, magnitudes and directions. Is object-on-string a force on the object?

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice ... . This supports ... because ... .

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg book rests on a table while a hand pushes down with 15 N.
Use g=10 N/kg.
Find the normal force and draw the third-law partner of each force on its separate object.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

Marzano check - circle: 1 2 3